MID-STATE TECHNICAL COLLEGE · IT PROGRAMS

Backup Strategies and Data Protection

Unit 10 · Virtualization and Software Defined Data Center (10-150-130)

MID-STATE TECHNICAL COLLEGE

Learning Outcomes

By the end of this unit, you will be able to:



Design a backup strategy

Choose among full, incremental, differential, and snapshot protection and apply the 3-2-1-1-0 rule.



Define recovery objectives

Set RTO, RPO, MTD, and WRT from a business impact analysis and weigh them against cost.



Configure automated protection

Schedule vzdump jobs and run Proxmox Backup Server with retention, verification, and garbage collection.



Plan for disaster recovery

Select recovery sites, test methods, and communication plans that support business continuity.

LESSON 1

Backup Fundamentals and Strategy Design

Why Data Gets Lost

Every backup strategy starts from the failures it has to survive:

- **Hardware failure** — a disk, controller, or entire host dies; storage pools degrade or go offline.
- **Human error** — the wrong VM deleted, a bad update applied, a config file overwritten.
- **Ransomware and sabotage** — attackers encrypt production data and hunt for the backups too.
- **Site-level disaster** — fire, flood, tornado, or extended power loss takes out the whole room.



Redundancy is not a backup

RAID, ZFS mirrors, and HA clusters keep a workload running through a hardware fault — but they copy a deletion or an encryption just as faithfully. A **backup** is a separate point-in-time copy you can go back to.

Four Ways to Protect Data

Each method trades storage and backup window against restore speed and complexity.

Method	What it captures	Restore needs	Trade-off
Full	Every block, every time	One backup set	Fastest restore; most storage and time
Incremental	Changes since the last backup of any type	Last full + every incremental	Smallest and fastest to take; longest chain
Differential	Changes since the last full	Last full + latest differential	Middle ground; grows daily until next full
Snapshot	Point-in-time state on the same storage	The original storage	Instant rollback; not an independent copy

Modern deduplicating systems blur these lines — every backup can be "incremental to take, full to restore."

Incremental vs. Differential Restores

Incremental chain

- Full Sunday, then a small incremental each night
- Thursday restore = **Sunday + Mon + Tue + Wed + Thu**
- One damaged link breaks every restore point after it
- Best when the backup window or bandwidth is tight

Differential chain

- Full Sunday, then a growing differential each night
- Thursday restore = **Sunday + Thu**
- Only two sets to read, so restores are faster and safer
- Each night copies more data than the night before

Incremental-forever: deduplicating targets send only changed chunks but index every snapshot as a complete image — no chain to replay at restore time.

Snapshots Are Not Backups

A snapshot freezes a VM's state on its own storage — perfect for undo, useless when that storage fails.

```
qm snapshot 101 pre-patch --description "Before updates"
qm listsnapshot 101
qm rollback 101 pre-patch
```

- Use snapshots for **short-term rollback** — before updates, upgrades, or lab experiments.
- They depend on the **same storage** as the VM: lose the pool or delete the VM and they are gone.
- Long snapshot chains cost performance; delete them once the change is confirmed good.

From 3-2-1 to 3-2-1-1-0

The classic rule still applies — ransomware added two requirements on top of it.

Digit	Requirement	Defends against
3	Three copies of the data — production plus two backups	A single copy failing or being deleted
2	Two different media or platforms	One storage technology failing the same way twice
1	One copy offsite	Fire, flood, theft, or site-wide outage
1	One copy immutable or air-gapped	Ransomware and insiders deleting the backups
0	Zero errors — backups verified and test-restored	Discovering a bad backup during the emergency

The rule says how many copies and where; your RPO says how often they are made.

Scheduling and Retention

Automation removes the human from the routine; retention decides how far back you can go.

- **Schedule** backups inside the quietest window and stagger large jobs so they do not collide.
- **Retention** stacks rules — `keep-last`, `keep-daily`, `keep-weekly`, `keep-monthly`, `keep-yearly`.
- **Tiering** keeps recent restore points on fast storage and moves history to cheaper or offsite media.
- Compliance can set a **minimum** retention; storage cost usually sets the maximum.



Example policy

```
keep-last=3
keep-daily=7
keep-weekly=4
keep-monthly=6
```

About 20 restore points spanning half a year.



LESSON 2

Recovery Objectives and Business Impact

RTO and RPO

Two numbers drive nearly every backup and DR design decision:

- **Recovery Point Objective (RPO)** — the most data you can afford to lose, measured back from the outage to the last good copy.
- **Recovery Time Objective (RTO)** — the longest the service can be down before it must be running again.
- A nightly backup means an RPO of up to **24 hours** — a full day of work is at risk.
- RTO includes detection, decision, restore, and boot — not just the restore copy time.



Rule of thumb

RPO sets how often you back up or replicate.

RTO sets how fast your restore technology and process must be.

Four Recovery Metrics

RTO and RPO are the headline numbers; MTD and WRT complete the recovery budget.

Metric	Definition	Drives
RTO · Recovery Time Objective	Maximum time to restore the system after a disruption	Restore technology, automation, staffing
RPO · Recovery Point Objective	Maximum acceptable data loss, measured in time	Backup frequency, replication, storage speed
MTD · Maximum Tolerable Downtime	Longest the business survives without the process	DR site choice and failover capability
WRT · Work Recovery Time	Time to validate data and resume normal work after restore	Runbooks, training, verification steps

Industry terms vary slightly between frameworks; NIST SP 800-34 uses MTD, RTO, and RPO.

Fitting Inside the MTD

The technical restore and the human recovery together must finish before the business limit:

$$RTO + WRT \leq MTD$$

- Start from the **MTD** the business gives you, then subtract the work recovery time.
- What remains is the **largest RTO** your design is allowed to have.
- If a test restore takes longer than that, the design — not the paperwork — has to change.

Worked example

MTD for order entry: **8 hours**

WRT to reconcile orders: **2 hours**

Required RTO: **6 hours or less**

Business Impact Analysis

A BIA turns "everything is critical" into ranked, defensible recovery targets.

- **Identify** the business processes and the systems each one depends on.
- **Quantify** the cost of downtime per hour — revenue, penalties, safety, reputation.
- **Map dependencies** — the web app is useless without DNS, AD, and its database.
- **Assign** MTD, RTO, and RPO per application, then rank recovery order.



Ask the business

Evaluate each application separately and gather input from its owners. When the answer is uncertain, err toward faster recovery and less data loss.

Tighter Objectives Cost More

Every step toward zero downtime and zero data loss adds infrastructure, licensing, and staff time.

Protection approach	Typical RPO	Typical RTO	Relative cost
Nightly backup to local storage	24 hours	Hours	Low
Frequent PBS backups + offsite sync	1–4 hours	1–4 hours	Moderate
Storage replication to a second site	Minutes	Under 1 hour	High
HA cluster + synchronous replication	Near zero	Minutes	Highest

HA is not backup: replication and failover copy corruption and deletions instantly. Even the top tier still needs point-in-time backups.

LESSON 3

Proxmox Backup Tools: vzdump and PBS



vzdump Backup Modes

vzdump offers three modes — each trades downtime against consistency.

Mode	Guest downtime	Consistency	Use when
`snapshot`	None (live)	Crash-consistent; better with guest agent	Default for nearly all VMs and CTs
`suspend`	Brief pause	Higher; guest paused during copy	CTs on storage without snapshots
`stop`	Shutdown to restart	Highest; clean shutdown first	Strict consistency, downtime allowed

Guest agent matters

With `agent: 1` set and the agent running, snapshot mode calls `guest-fsfreeze-freeze` and `guest-fsfreeze-thaw` around the copy.

vzdump from the Command Line

The GUI and scheduled jobs call the same tool you can run by hand:

```
vzdump 101 --mode snapshot --compress zstd --storage <backup-storage>
vzdump --all --mode snapshot --storage <backup-storage>
qmrestore <backup-file> 201 --storage <target-storage>
pct restore 301 <backup-file> --storage <target-storage>
```

- File storage gets archives like `vzdump-qemu-101-<timestamp>.vma.zst`; containers produce `.tar.zst`.
- Restore to a **new VMID** to test a backup without touching production.

Scheduled Backup Jobs

Datacenter → Backup turns one-off commands into repeatable, monitored jobs.

- **Schedule** with calendar events — `mon..fri 21:00`, `sat 02:00`, `*-*-01 03:00`.
- **Selection** by VMID, by pool, or all guests with exclusions.
- **Retention** per job using the same `keep-*` options from Lesson 1.
- **Notifications** on failure, so a broken job is noticed the next morning.



Backup fleecing

Advanced tab option that caches blocks on fast local storage during the backup, so a slow target cannot stall the VM's disk I/O. Example: `--fleecing enabled=1,storage=local-lvm`.

Proxmox Backup Server Architecture

A dedicated backup appliance built around deduplicated, verifiable chunk storage.

- **Chunk store** — backups split into chunks named by their SHA-256 hash; identical chunks stored once.
- **Incremental transfers** — dirty bitmaps let running VMs send only changed blocks.
- **Every snapshot is full** — restore any point directly, with no chain to replay.
- **Granular restore** — browse and pull individual files from VM and CT backups.



Connecting PVE to PBS

Add the datastore as storage type `pbs` in Datacenter → Storage, using the server address, datastore name, credentials, and the PBS certificate **fingerprint**. Its web UI listens on `8007`.

What's New in PBS 4.x

The current release line adds cloud-ready storage and makes offsite copies easier.

Feature	Why it matters
S3-compatible datastores	Back up to object storage; preview in 4.0, fully supported as of 4.2
Removable datastores · run-on-mount	A sync job starts automatically when a rotation drive is plugged in
Parallel sync workers	Sync jobs process multiple groups at once to improve throughput on slow links
Move groups and namespaces	Reorganize backups inside a datastore without re-creating them

PBS 4 runs on Debian 13 "Trixie" and accepts backups from both PVE 8 and PVE 9 clients.

Client-Side Encryption

Encrypt before data leaves the host — and guard the key.

- Proxmox VE encrypts backups **client-side** with AES-256-GCM before sending them to PBS.
- The backup server stores only **ciphertext** — useful for offsite and cloud datastores.
- Keep the **encryption key** outside the cluster; a paper key or master key is the recovery path.



SECURITY FOCUS

If the only copy of the key lives on the Proxmox host and that host is lost, every encrypted backup is lost with it.

Why it matters: Lost keys make backups permanently unreadable — store and test key recovery offline.

Verify, Prune, and Garbage Collection

Three scheduled jobs keep a PBS datastore trustworthy and right-sized.

Job	What it does	Schedule tip
Verify	Re-reads chunks and checks their SHA-256 hashes against the index	Weekly; re-verify old snapshots periodically
Prune	Removes snapshot indexes that fall outside the retention rules	Daily, after backups finish
Garbage collection	Marks chunks still in use, then deletes unreferenced ones	After pruning; this is what frees disk space

Protected snapshots: mark a backup as protected and prune will skip it — useful for pre-upgrade or legal-hold restore points.

Getting Copies Offsite with PBS

Online offsite copies

- **Sync jobs** replicate a datastore to a second PBS, pull or push
- **S3 datastores** send backups to cloud or on-prem object storage
- Separate credentials limit what a compromised host can delete

Offline and air-gapped copies

- **Removable datastores** on rotating USB drives
- **Tape backup** to LTO media stored away from the site
- Disconnected media cannot be encrypted by malware on the network

Map it: local PBS datastore = copy two; synced or S3 copy = offsite; removable or tape = the immutable "1"; verify jobs = the "0".

LESSON 4

Disaster Recovery and Business Continuity

Continuity vs. Recovery

Business Continuity (BC)

- Keeps critical business functions running during a disruption
- Covers people, facilities, suppliers, and manual workarounds
- Owned by leadership; guided by ISO 22301

Disaster Recovery (DR)

- Restores IT systems and data after a disruption
- Covers backups, recovery sites, runbooks, and failover
- Owned by IT; guided by NIST SP 800-34

Relationship: DR is one part of BC — the BIA from Lesson 2 feeds both, and both must be tested together.

Recovery Site Options

The faster the required recovery, the more of the site must already be running.

Site	What is ready	Typical RTO	Cost profile
Hot	Running hardware with current, replicated data	Minutes	Highest — a second data center
Warm	Hardware in place; data restored from recent backups	Hours	Moderate
Cold	Space, power, and network only	Days to weeks	Lowest ongoing cost
Cloud DR	Replicated data; compute started on demand	Hours	Pay-as-you-go; low idle cost

A second Proxmox cluster receiving PBS sync jobs is a practical warm site for small and mid-sized organizations.

The DR Planning Cycle

Disaster recovery planning is a loop that repeats every time the environment changes:



Risk assessment and BIA

Identify threats and vulnerabilities and quantify their business impact.



Recovery strategy selection

Choose backup methods and recovery sites that meet each system's RTO and RPO within budget.



Plan development and documentation

Write runbooks, contact lists, and decision authority so recovery works without the usual experts.



Training and testing

Exercise the plan, train staff, and feed lessons learned back into the next risk assessment.

Testing the Plan

Test methods range from discussion to a real outage — each step adds confidence and risk.

Method	What happens	Disruption
Tabletop exercise	Team talks through a scenario step by step using the plan	None
Walkthrough	Each owner reviews their procedures in detail for gaps	None
Simulation	Recovery staff act out a realistic scenario with real tools	Low
Parallel test	Systems restored at the recovery site while production keeps running	Low to moderate
Full interruption	Production is shut down and the recovery site takes over	High

Without testing, you are guessing whether the recovery objectives can actually be met.

Communication and Emerging Threats

Recovery depends on people being able to coordinate when normal systems are gone.

- **Defined roles** — who declares a disaster, who restores, who talks to staff and customers.
- **Out-of-band contact** — phone trees and printed lists that work without email or Teams.
- **Decision authority** written down in advance, so stress does not stall recovery.

Evolving threats: severe weather, ransomware aimed at backup systems, and slow hardware supply chains all mean plans must adapt — review them at least annually.

Security Focus: Protecting the Backups

SECURITY FOCUS

Treat the backup system as critical infrastructure: separate admin credentials from the production domain, give Proxmox VE a backup-only token on PBS, encrypt client-side, keep at least one immutable or offline copy, and verify restores on a schedule.

Why it matters: Ransomware operators delete or encrypt backups first — if they can reach every copy, you will be negotiating instead of restoring.

- Least privilege: a PVE host should be able to **write backups**, not delete the datastore.
- Isolate the backup network and management interface from user and guest traffic.

Hands-On Labs: Backup and PBS

NETLAB lab series · your assigned POD · details and due dates in Blackboard Ultra



Lab 10 · Backup Solutions

- Create, list, and roll back VM snapshots
- Configure vzdump backups and storage
- Schedule automated backup jobs
- Restore a VM and verify it boots

Lab 11 · Proxmox Backup Server

- Deploy PBS and create a datastore
- Connect Proxmox VE and run backups
- Configure pruning, GC, and verify jobs
- Perform a full and a file-level restore

Graded against observable outcomes on the live systems — a working configuration, not a screenshot.

Standards and Resources

Standards and guidance

- **ISO 22301:2019** — business continuity management
- **NIST SP 800-34** — IT contingency planning
- **CISA #StopRansomware Guide** — offline, tested backups
- **DRI and BCI** professional practices

Documentation

- `pve.proxmox.com/pve-docs/chapter-vzdump.html`
- `pbs.proxmox.com/docs/`
- `nist.gov/cyberframework`
- `thebci.org`

Product versions referenced: Proxmox VE 9.2 and Proxmox Backup Server 4.2 (current as of Fall 2026).

Key Takeaways

- Redundancy keeps you running; only independent, point-in-time backups let you go back. Follow 3-2-1-1-0.
- RPO sets backup frequency, RTO sets restore speed — and RTO + WRT must fit inside the MTD.
- vzdump snapshot mode plus Proxmox Backup Server gives deduplicated, encrypted, verifiable backups.
- Protect the backups themselves: separate credentials, offline copies, and safely stored encryption keys.
- A DR plan is only as good as its last test — restore regularly and keep the plan current.

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Questions?

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